

Introduction to Aerospace Engineering

Lecture slides

Introduction to Aerospace Engineering

Aerodynamics 1&2

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Don Schir 091

1 & 2.

Fundamentals of aerodynamics

Anderson chapters 2.1 – 2.5 and 4.1 – 4.4

Leonard Euler Daniel Bernoulli



1707-1783



1700-1782



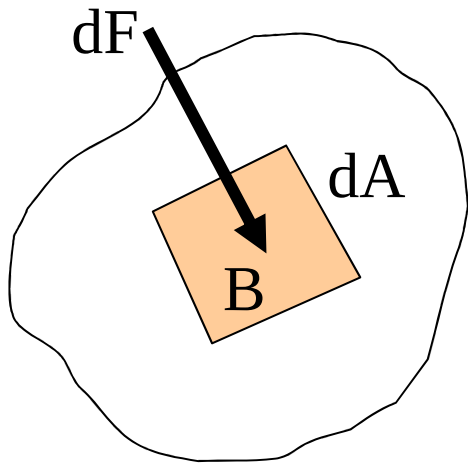
Subjects

Fundamental quantities

- pressure
- density
- temperature
- velocity

Equation of state

Pressure is the normal force per unit area on a surface



dA = element surface around B

dF = force on 1 side of dA

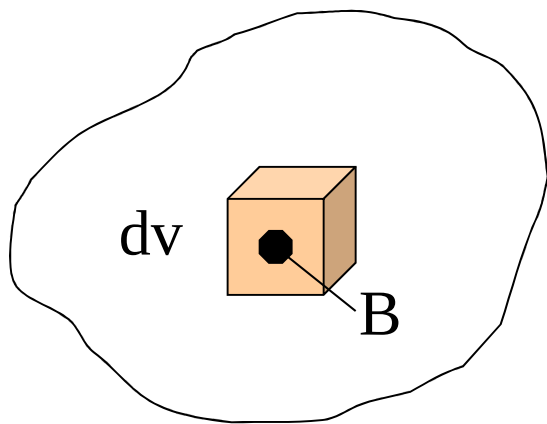
The pressure in point B is:

$$p = \lim \left(\frac{dF}{dA} \right), dA \rightarrow 0$$

Pressure is force per unit area

Dimension of pressure = $[N/m^2]$

Density of a substance is its mass per volume



dv = element volume around B

dm = mass of gas in dv

The density in point B is:

$$\rho = \lim \left(\frac{dm}{dv} \right), dv \rightarrow 0$$

Density is mass per volume

Dimension of density = $[\text{kg}/\text{m}^3]$

Temperature is a measure of the average kinetic energy of the particles in the gas

Temperature is a measure of the average kinetic energy of the particles in the gas:

$$KE = \frac{3}{2}kT,$$

with k = Boltzmann constant (1.38×10^{-23} J/K)

Dimension of temperature : K (Kelvin)
°C (degree Celsius),

with $K = 273.15 + C$

Relation among pressure, density and temperature is the equation of state

A ***perfect gas*** is a gas in which intermolecular forces are negligible

The equation of state for a perfect gas is:

$$p = \rho RT$$

With gas constant R :

$$R = 287,05 \text{ J/(kg)(K)}$$

The equation of state for a non-perfect or actual gas

For an actual gas the equation of state is approximated by the Berthelot equation:

$$\frac{p}{\rho RT} = 1 + \frac{ap}{T} - \frac{bp}{T^3}$$

The difference with the equation of state for a perfect gas becomes smaller as p decreases or T increases

(The distance between the molecules increases)

The *Benedict-Webb-Rubin* (BWR) equation of state [2,3] within temperature range from -30 to 150°C, for densities up to 900kg/m³, and maximum pressure of 200bar:

$$p = \frac{RT}{v} + \left(B_0 RT - A_0 - \frac{C_0}{T^2} \right) \left(\frac{1}{v} \right)^2 + (bRT - a) \left(\frac{1}{v} \right)^3 + a\alpha \left(\frac{1}{v} \right)^6 + \frac{c}{T^2} \left(\frac{1}{v} \right)^3 \left[1 + \gamma \left(\frac{1}{v} \right)^2 \right] \exp \left[-\gamma \left(\frac{1}{v} \right)^2 \right]$$

where

$A_0, B_0, C_0, a, b, c, \alpha, \gamma$ — gas constants for the equation

$$A_0 = 0.313 \text{ kg m}^5 \text{ s}^{-2} \text{ mol}^{-2}$$

$$B_0 = 5.1953 \times 10^{-5} \text{ m}^3 \text{ mol}^{-1}$$

$$C_0 = 1.289 \times 10^4 \text{ kg m}^5 \text{ K}^2 \text{ s}^{-2} \text{ mol}^{-2}$$

$$a = 1.109 \times 10^{-5} \text{ kg m}^8 \text{ s}^{-2} \text{ mol}^{-3}$$

$$b = 3.775 \times 10^{-9} \text{ m}^6 \text{ mol}^{-2}$$

$$c = 1.398 \text{ kg m}^8 \text{ K}^2 \text{ s}^{-2} \text{ mol}^{-3}$$

$$\alpha = 9.377 \times 10^{-14} \text{ m}^9 \text{ mol}^{-3}$$

$$\gamma = 5.301 \times 10^{-9} \text{ m}^6 \text{ mol}^{-2}$$



Ex 2.1 Compute the temperature in a point on a wing of a Boeing 747, where pressure and density are given to be: $0.7 \times 10^5 \text{ N/m}^2$ and 0.91 kg/m^3

$$\begin{aligned} T &= 268.0 \text{ K} \\ &= -5.2 \text{ }^\circ\text{C} \end{aligned}$$

Pressure, density and temperature under standard conditions

$$P_s = 1.01325 * 10^5 \text{ N/m}^2$$

$$\rho_s = 1.225 \text{ kg/m}^3$$

$$T_s = 288.15 \text{ K}$$



Ex. Compute the total weight of air in a room of
30 m x 15 m x 5 m under standard atmospheric
conditions at sea level.

$$\text{Volume} = 2250 \text{ m}^3$$

$$\rho = 1.225 \text{ kg/m}^3$$

$$\text{Weight} = 2756 \text{ kg}$$

Definition of specific volume

$$v = \frac{1}{\rho}$$

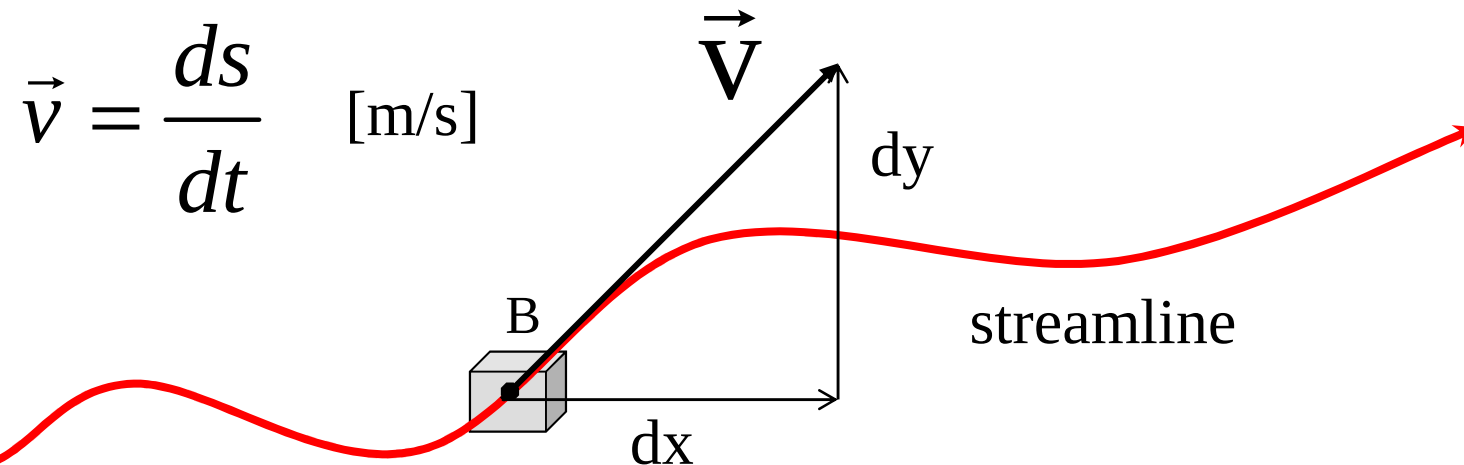
$$pv = RT$$



Ex 2.3 Compute the density and specific volume of air in a wind tunnel at $P = 0.3 \text{ atm}$, and -100°C .

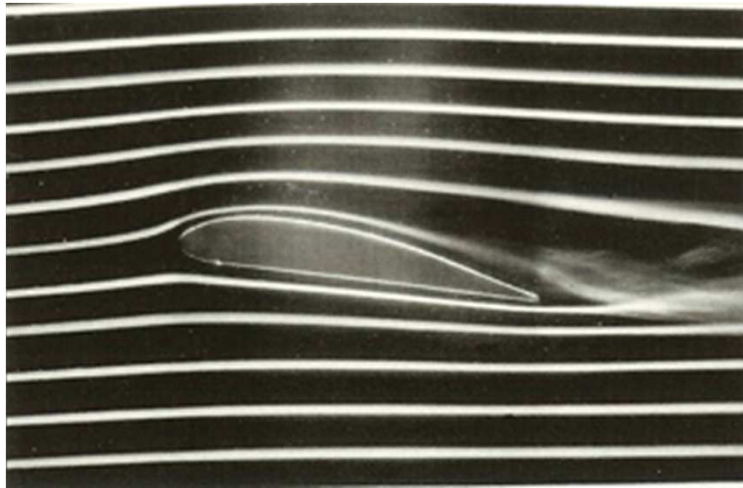
$$\rho = \frac{p}{RT} \quad \begin{aligned} \rho &= 0.3 \times 10^5 / (287.05 \times 173.15) \\ &= 0.60 \text{ kg/m}^3 \\ v &= 1.66 \text{ m}^3/\text{kg} \end{aligned}$$

Velocity and streamlines



The velocity in point B is the velocity of an infinitesimally small fluid element as it sweeps through B

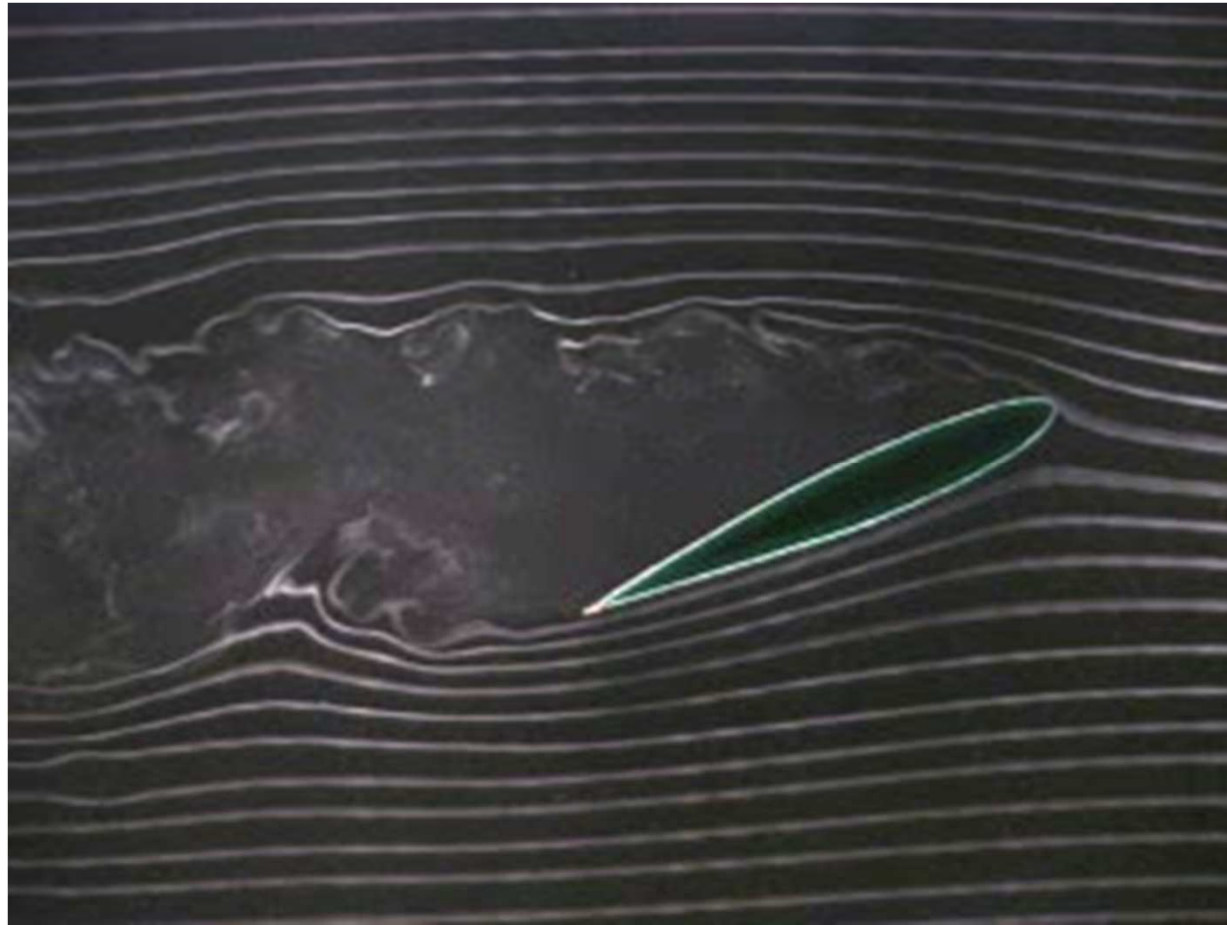
Visualisation of the velocity and streamlines



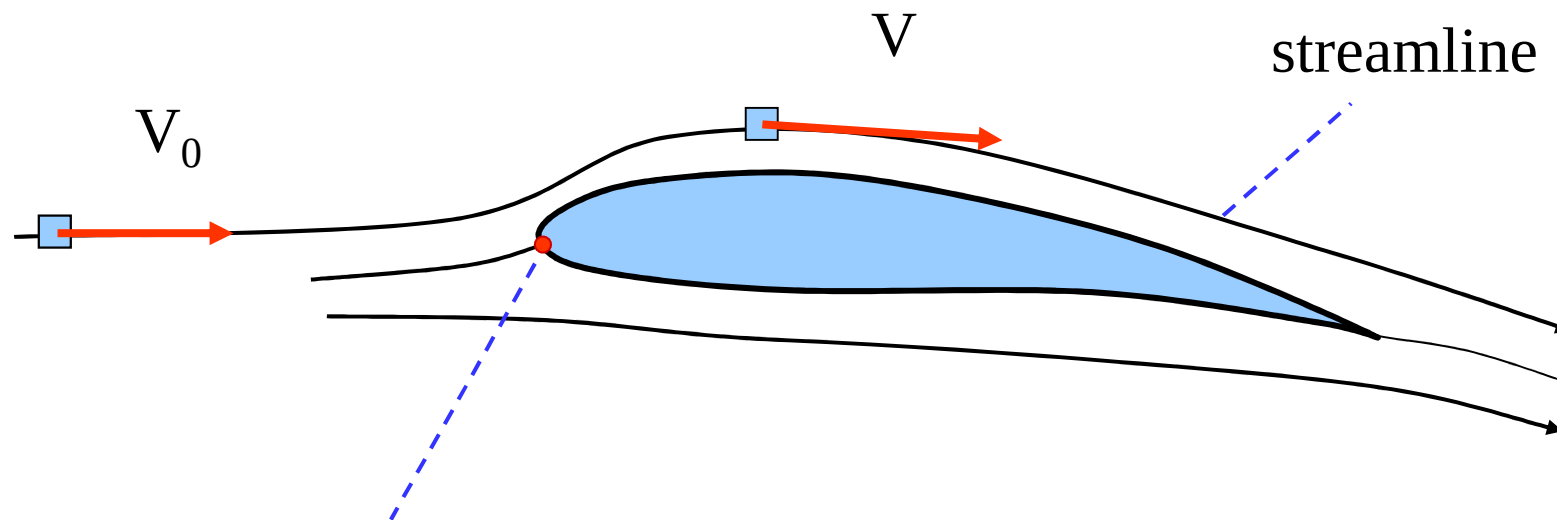
Smoke traces in air

Aluminum Particles in water





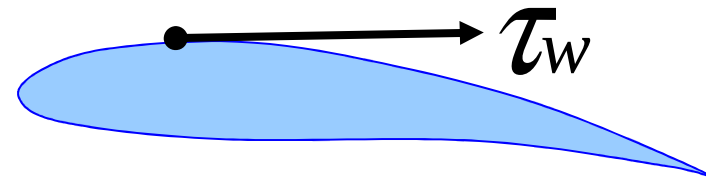
Velocity and streamlines



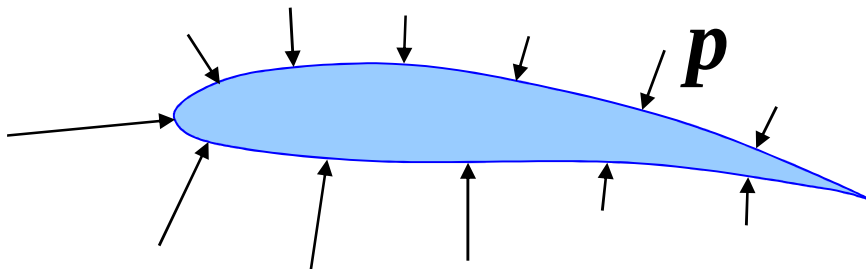
Stagnation point (stuwpunt, $V = 0$)

Aerodynamic forces

Shear stress or friction force



Pressure distribution





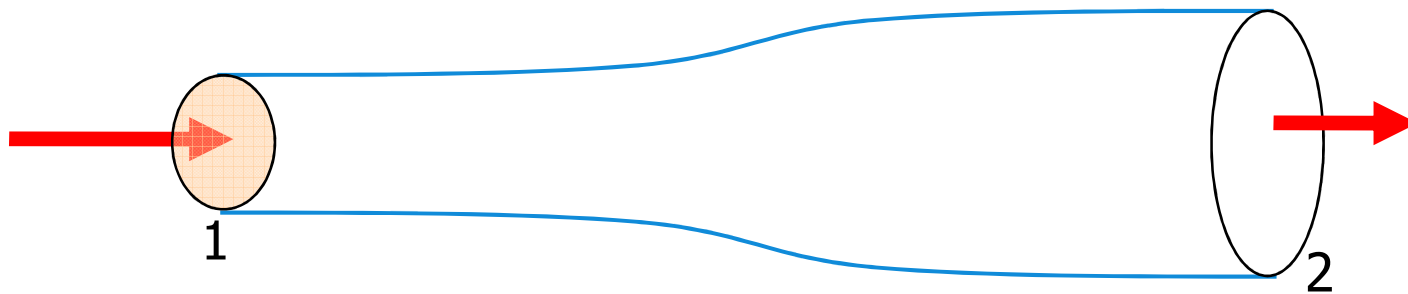
Homework:

Problems 2.1, 2.7 and 2.8 from
Anderson, page 102 (sixth edition)

Fundamental equations

- **Continuity equation:** Conservation of mass
- **Momentum equation (Euler eqn.):** Conservation of momentum
- **Bernoulli's law for** incompressible flow

Continuity equation: Time rate of change of the mass of a material region is zero (mass is conserved)



For 1-directional
incompressible flow: $\dot{m}_{\text{in}} = \dot{m}_{\text{out}}$


$$\dot{m} = \frac{dm}{dt} \quad dm = \rho (Qdt), \quad Q = AV$$

$$\dot{m} = \frac{dm}{dt} = \frac{\rho AV dt}{dt} = \rho AV$$

$$\rho AV = \text{constant}$$

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2$$

Continuity equation for 1-dimensional incompressible flow

(assumption: ρ and V uniformly distributed over A , or:

ρ and V are *mean* values)

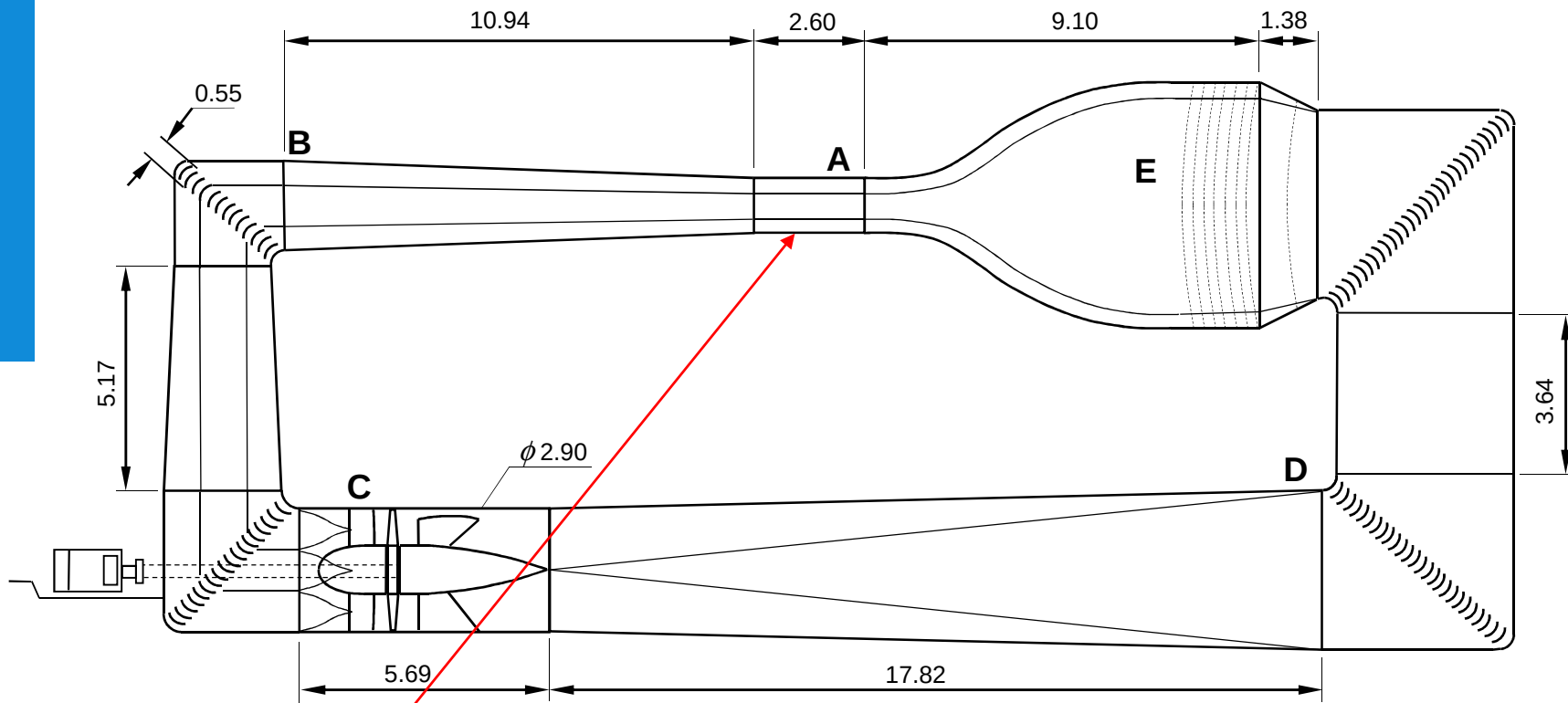
Example 4.1: Consider a convergent duct with an inlet area $A_1 = 5 \text{ [m}^2\text{]}$. Air enters this duct with a velocity $V_1 = 10 \text{ [m/s]}$ and leaves the duct exit with a velocity $V_2 = 30 \text{ [m/s]}$. What is the area of the duct exit?

Assume a steady, incompressible 1-directional flow

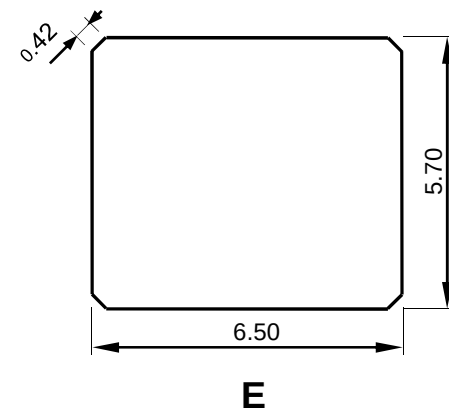
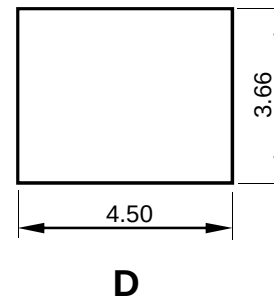
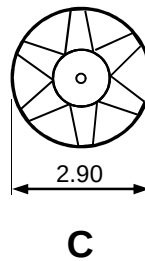
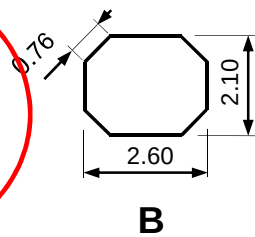
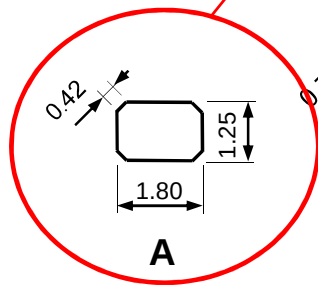
$$\rho A_1 V_1 = \rho A_2 V_2$$

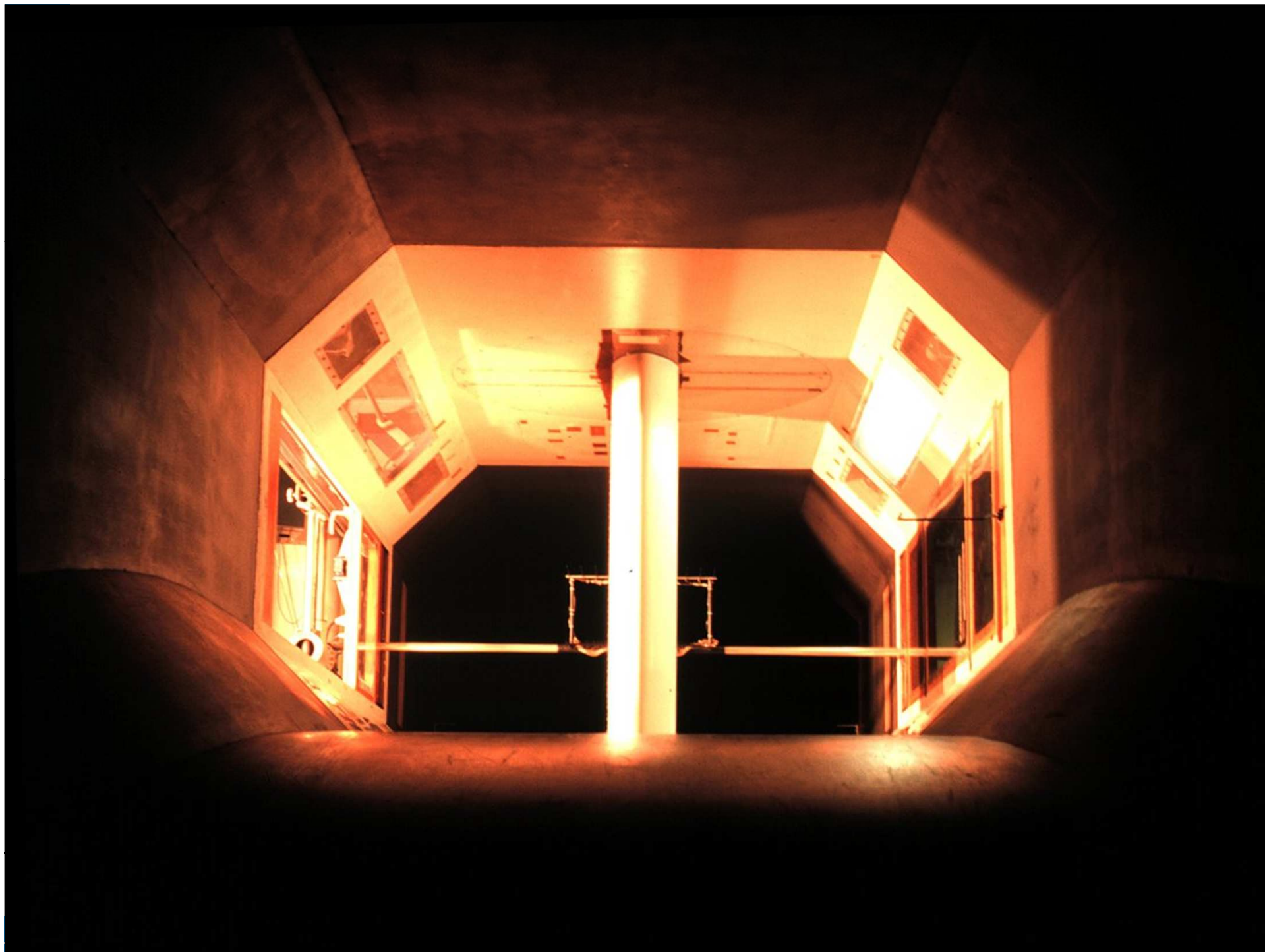
$$A_2 = \frac{\rho A_1 V_1}{\rho V_2} = A_1 \frac{V_1}{V_2} = 5 * \frac{10}{30} = 1.67 \text{ [m}^2\text{]}$$

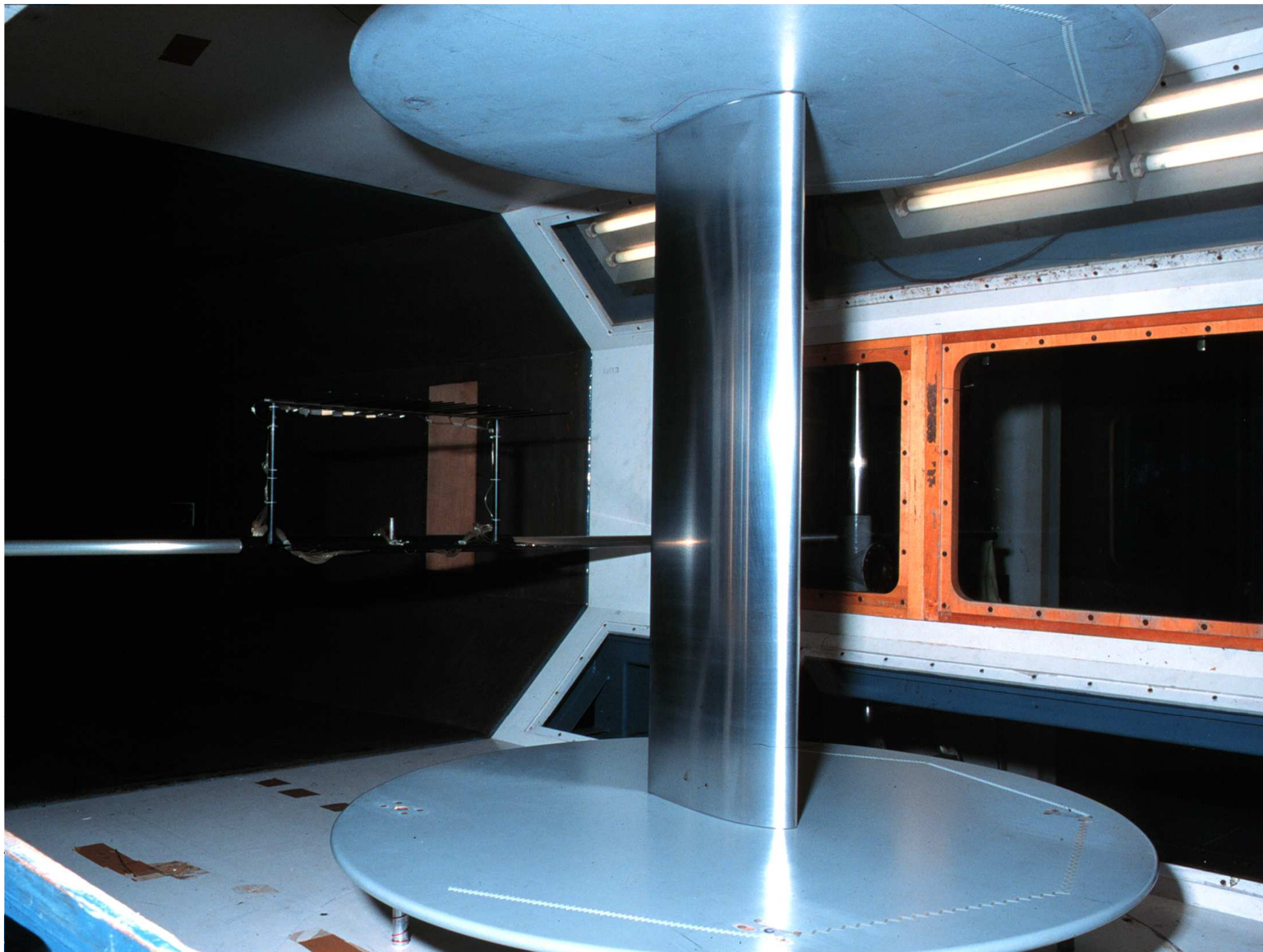
Application of the continuity law:..... e.g. wind tunnels

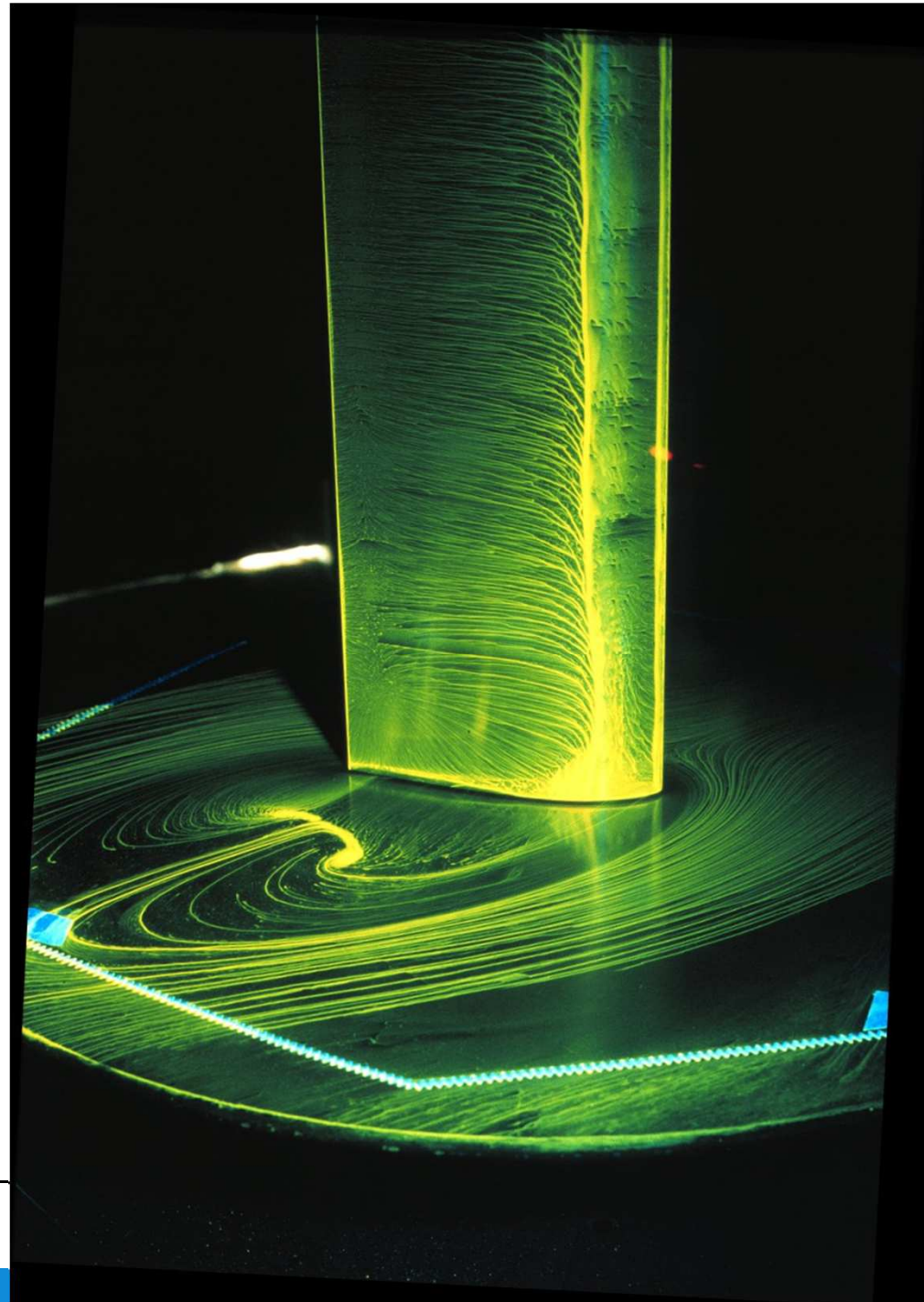


Sizes in meters

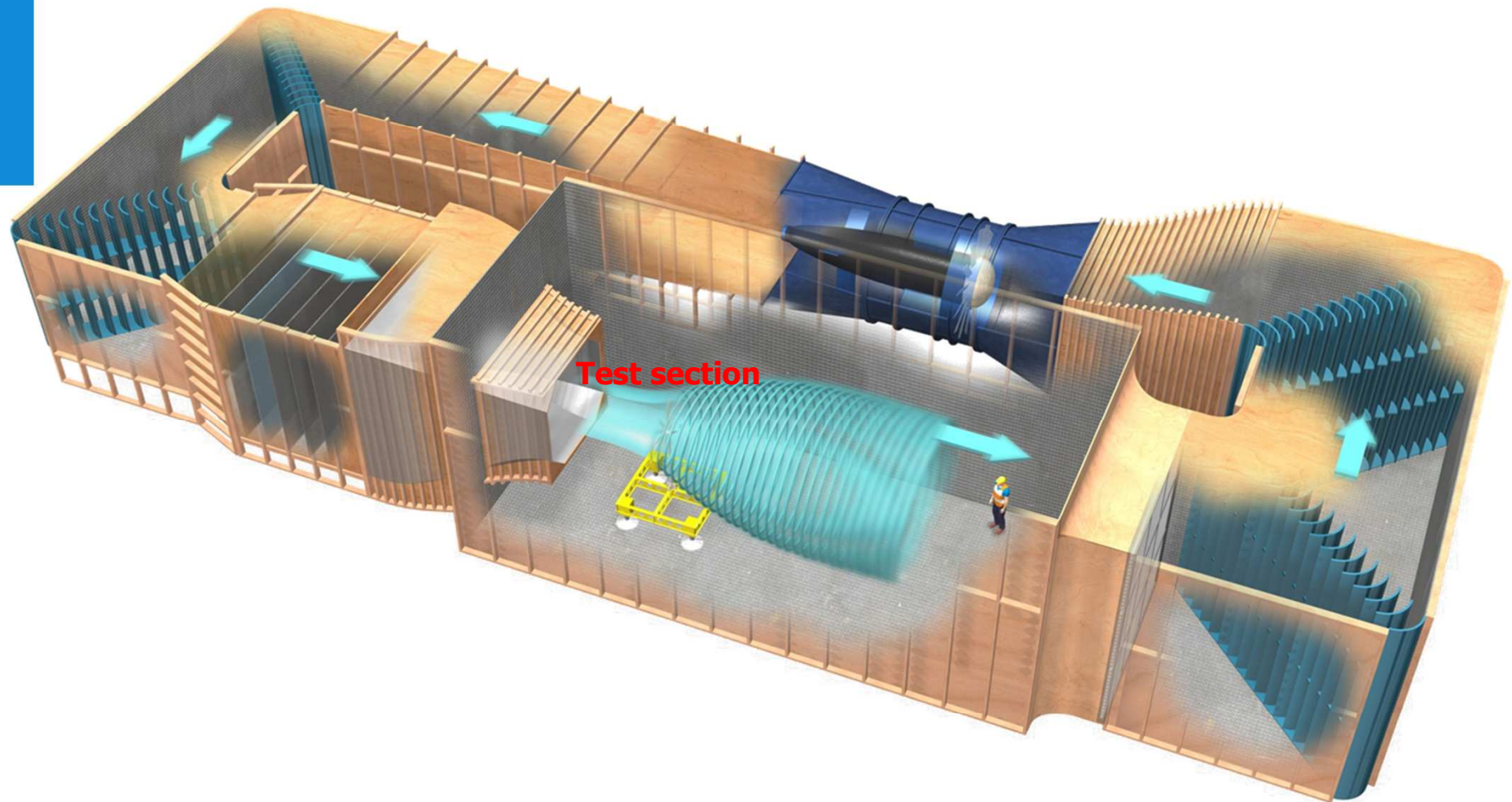


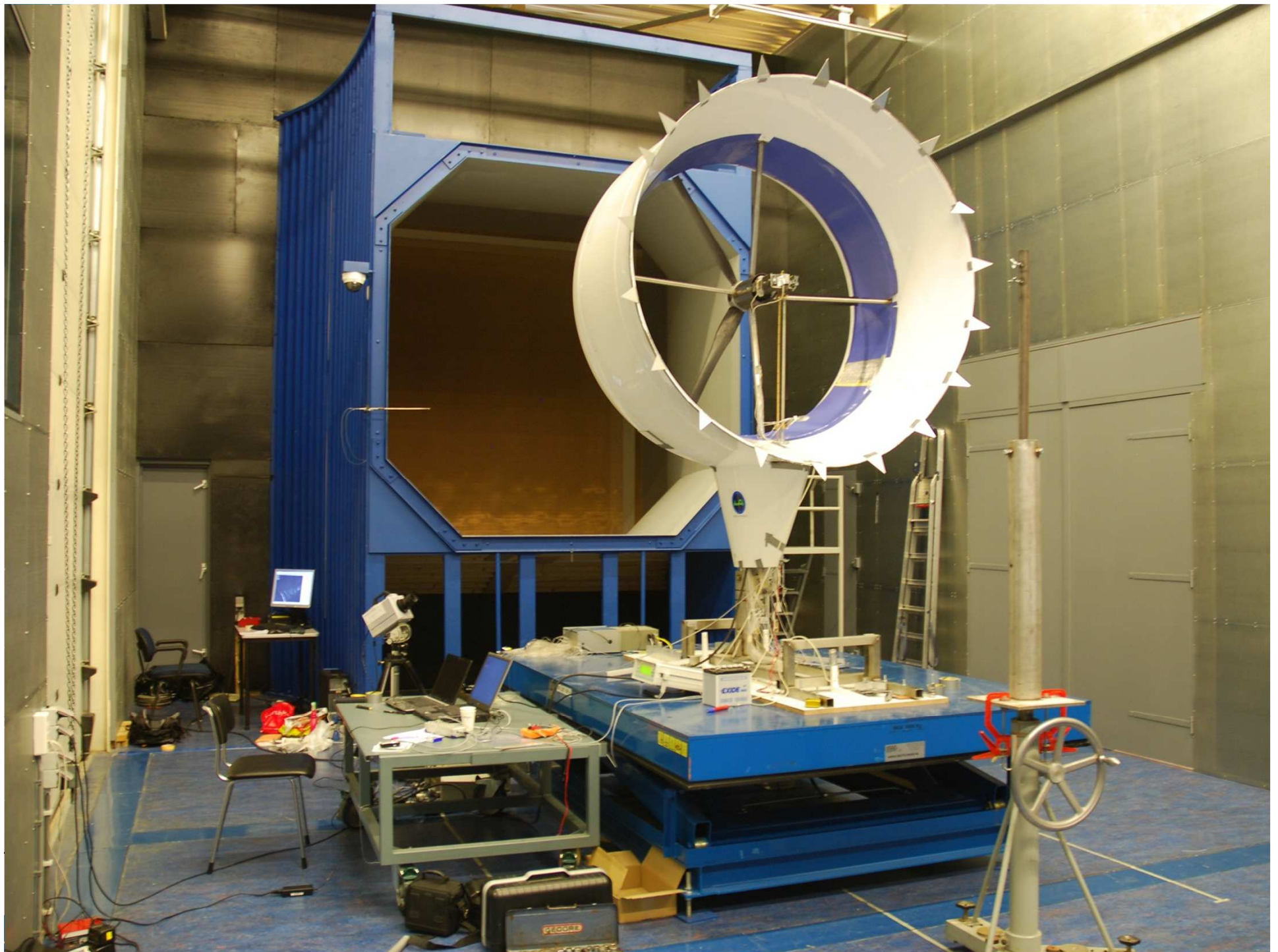




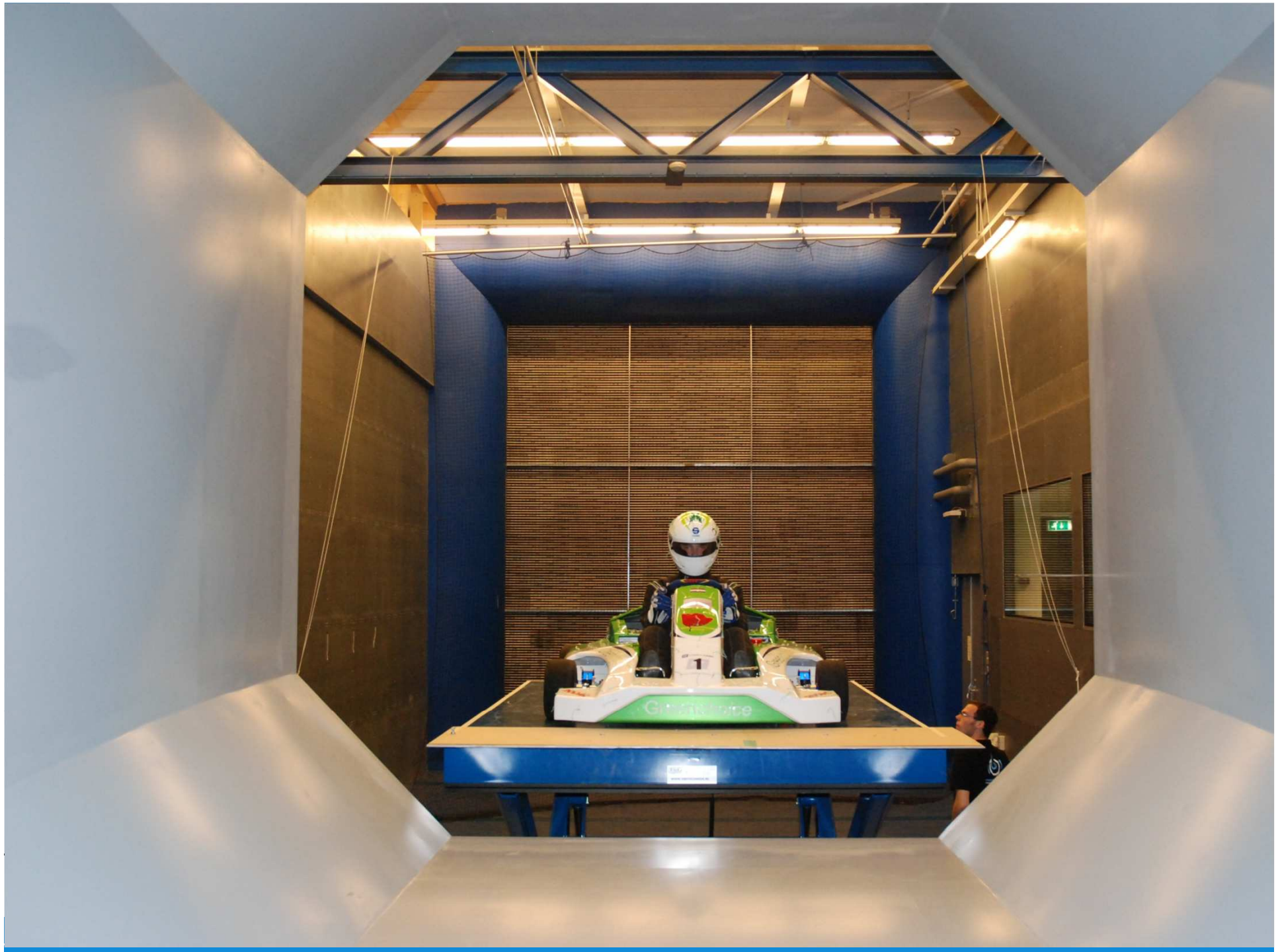


Circuit layout of Delft University Open-jet Facility







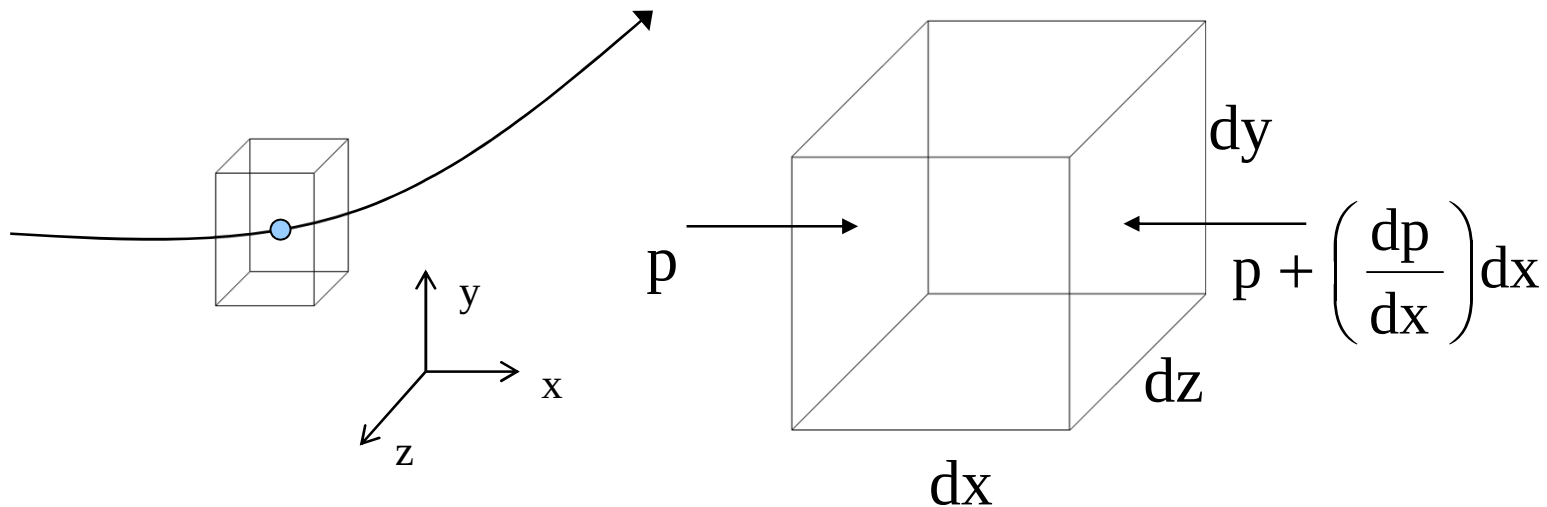


Euler equation

Newton's Second Law:

$$F = m.a \quad (\text{Force} = \text{mass} * \text{acceleration})$$

Applied to a flowing gas :



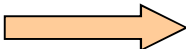
Euler equation, left hand part of

$$F = m \cdot a$$

In reality 3 forces act on this element:

- Pressure force
- Friction force
- Gravity force

In the derivation we neglect 2 forces:

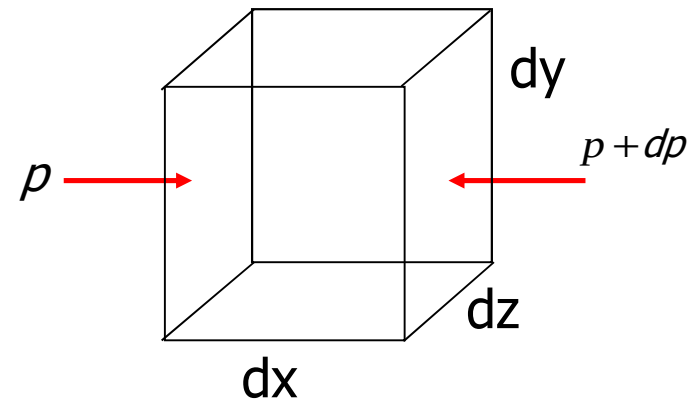
- Neglect the gravity force (small)
- Neglect the viscosity  No friction forces

Euler equation, left hand part

($F=m \cdot a$)

The force in **x-direction**:

$$F = p dydz - \left(p + \frac{dp}{dx} dx \right) dydz$$



Hence: $F = -\frac{dp}{dx} dxdydz$ = force on fluid element due to pressure

Euler equation, right hand part ($F=m*a$) and resulting equation

The mass (**m**) of the fluid element is : $m = \rho \cdot \text{Vol} = \rho dx dy dz$

Acceleration (**a**) of the fluid element is: $a = \frac{dV}{dt} = \frac{dV}{dx} \frac{dx}{dt} = \frac{dV}{dx} V$

Substitution in $F = m*a \longrightarrow -\frac{dp}{dx}(dx dy dz) = \rho(dx dy dz)V \frac{dV}{dx}$

Or: $dp = -\rho V dV$

Resulting equation:

$$dp = -\rho V dV$$

1. It is a relation between Force and Momentum
2. Also called the momentum equation

Euler equation



Euler (1707 - 1783)

Swiss Mathematician

$$e^{i\pi} + 1 = 0 \quad e^{i\phi} = \cos \phi + i \sin \phi$$

Euler equation

$$dp = -\rho V dV$$

Keep in mind for this equation:

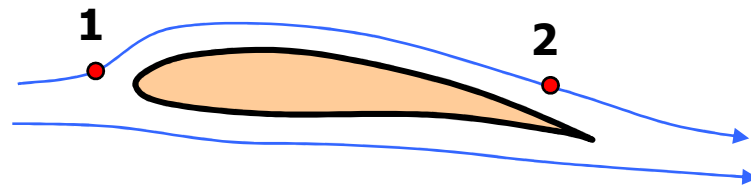
- Gravity forces are neglected
- Viscosity is neglected (inviscid flow)
- Steady flow
- Flow may be compressible!

The Euler equation is a *Differential equation*

Bernoulli's law for inviscid and incompressible flow, gravity forces neglected

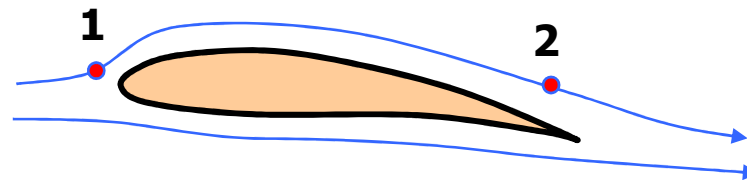
Integrate Euler equation ($dp + \rho V dV = 0$) along streamline between point 1 and 2:

$$\int_{p_1}^{p_2} dp + \int_{V_1}^{V_2} \rho V dV = 0$$



Bernoulli's law for inviscid and incompressible flow, gravity forces neglected

$$\int_{p_1}^{p_2} dp + \int_{V_1}^{V_2} \rho V dV = 0 \rightarrow (p_2 - p_1) + \rho \left(\frac{1}{2} V_2^2 - \frac{1}{2} V_1^2 \right) = 0$$



Bernoulli's law for inviscid and incompressible flow, gravity forces neglected

$$p_1 + \frac{1}{2} \rho V_1^2 = p_2 + \frac{1}{2} \rho V_2^2$$

$$\boxed{p} + \boxed{\frac{1}{2} \rho V^2} = p_t \text{ constant along a streamline}$$

↑ ↑ ↑

Static Dynamic Total

pressure pressure pressure

Bernoulli's law for inviscid and incompressible flow, gravity forces neglected

Bernoulli's law:

$$p + \frac{1}{2} \rho V^2 = \text{constant along a streamline}$$

Daniel Bernoulli



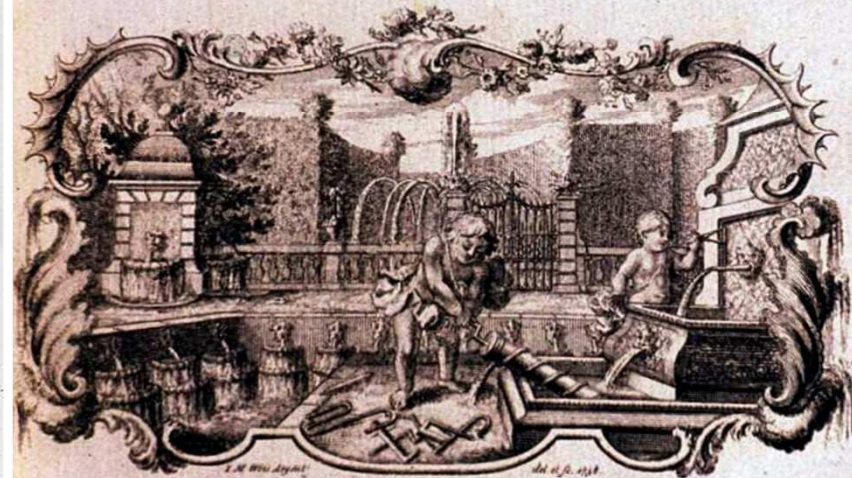
Daniel Bernoulli (1700-1782)
*Dutch born (Groningen, 29 jan. 1700),
Swiss Mathematician*

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 ACAD. SCIENT. IMPER. PETROPOLITANÆ, PRIUS MATHESIOS
 SUBLIMIORIS PROF. ORD. NUNC MEMBRI ET PROF. HONOR.
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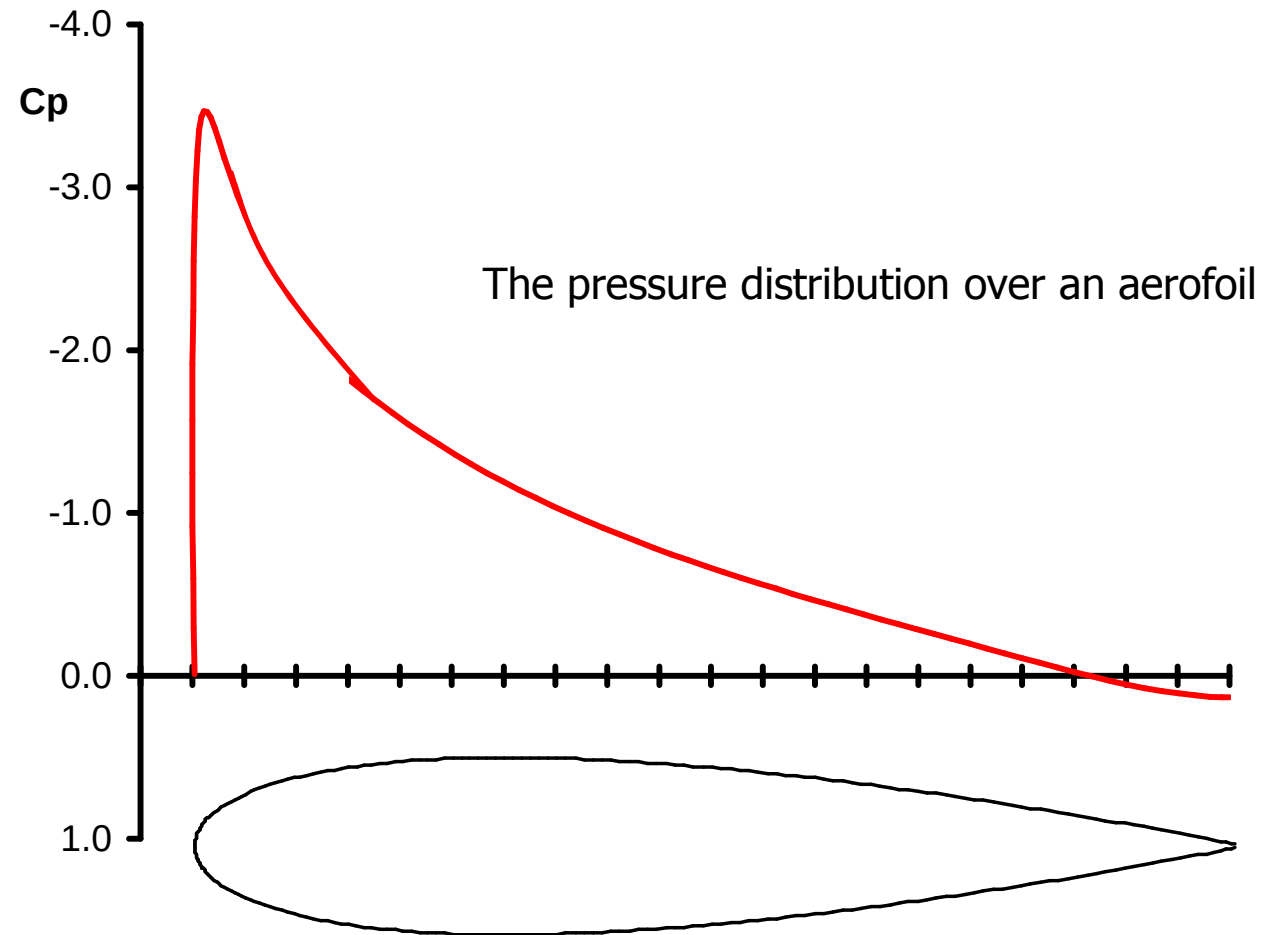


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1738



Application of Bernoulli's principle



Remember

- Bernoulli equation only for **inviscid = frictionless, incompressible** flow
- Bernoulli equation is valid **along a streamline**
- For **compressible** flow => Euler equation !
- Momentum equation, Euler equation and Bernoulli equation is in fact **$\mathbf{F} = \mathbf{m} \cdot \mathbf{a}$** (Newton) applied to fluid dynamics.

Ex 4.4. Consider the same convergent duct as in example 4.1. If the air pressure and temperature at the inlet are $p_1 = 1.2 \times 10^5 \text{ N/m}^2$ and $T_1 = 330\text{K}$, respectively, calculate the pressure at the exit.

Example: Consider the same convergent duct as in the previous example. If the air pressure and temperature at the inlet are $p_1 = 1.2 \times 10^5 \text{ [N/m}^2\text{]}$ and $T_1 = 330 \text{ [K]}$ respectively, calculate the pressure at the exit

- **Assume inviscid, incompressible flow**
- **Neglect the gravity forces**
- **On a streamline in the duct: Apply Bernoulli's law!**

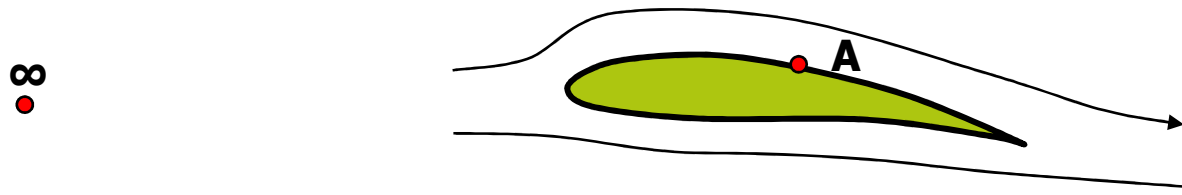
$$p_1 + \frac{1}{2} \rho V_1^2 = p_2 + \frac{1}{2} \rho V_2^2 \rightarrow p_2 = p_1 + \frac{1}{2} \rho (V_1^2 - V_2^2)$$

$$p_1 = \rho R T_1 \rightarrow \rho = \frac{p_1}{R T_1} = \frac{1.2 \times 10^5}{(287.15 \times 330)} = 1.266 \text{ [kg/m}^3\text{]}$$

$$p_2 = p_1 + \frac{1}{2} \rho (V_1^2 - V_2^2) = 1.2 \times 10^5 + \frac{1}{2} \times 1.266 \times (10^2 - 30^2) = 1.19 \times 10^5 \text{ [N/m}^2\text{]}$$

Example: Consider an airfoil in a flow of air, where far ahead (upstream) of the airfoil, the pressure, velocity and density are: $1.01 \times 10^5 \text{ [N/m}^2\text{]}$, 150 [km/h] and $1.225 \text{ [kg/m}^3\text{]}$ respectively. At a given point A on the airfoil the pressure is $9.95 \times 10^4 \text{ [N/m}^2\text{]}$. What is the velocity in point A?

- Assume inviscid, incompressible flow
- Neglect the gravity forces
- On a streamline near the airfoil: Apply Bernoulli's law!



Example: Consider an airfoil in a flow of air, where far ahead (upstream) of the airfoil, the pressure, velocity and density are: 1.01×10^5 [N/m²], 150 [km/h] and 1.225 [kg/m³] respectively. At a given point A on the airfoil the pressure is 9.95×10^4 [N/m²]. What is the velocity in point A?

Assume that point A and ∞ are on the same streamline. Bernoulli then gives:

$$p_{\infty} + \frac{1}{2} \rho_{\infty} V_{\infty}^2 = p_A + \frac{1}{2} \rho_{\infty} V_A^2 \rightarrow V_A = \sqrt{\frac{(p_{\infty} - p_A) + \frac{1}{2} \rho_{\infty} V_{\infty}^2}{\frac{1}{2} \rho_{\infty}}}$$
$$V_A = \sqrt{\frac{2(p_{\infty} - p_A)}{\rho_{\infty}} + V_{\infty}^2} = \sqrt{\frac{2(1.01 \times 10^5 - 9.95 \times 10^4)}{1.225} + \left(\frac{150}{3.6}\right)^2} = 64.7 \text{ [m/s]}$$

- 
- Problems 4.3 and 4.5